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Comprehensive Molecular Profiling of AcrAB-TolC Efflux Pump Genes in *Salmonella typhi* Isolates from Typhoid Infected Patients

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Abstract

Salmonella typhi is a facultative anaerobic, rod-shaped, Gram-negative bacterium that causes typhoid fever, a potentially fatal systemic infection. This study aimed to characterize antibiotic susceptibility patterns, mutations at the molecular level, and efflux pump genes in clinical isolates. In this study, blood samples ($n = 2950$) were collected from suspected typhoid-infected patients, and 380 (12.88%) bacterial isolates were found, comprising 144 (37.89%) Gram-positive and 236 (62.10%) Gram-negative bacteria. *S. typhi* was identified in 95 isolates (25%), corresponding to an overall prevalence of 3.22%. Biochemical identification was performed by Analytical Profile Index (API) 20-E strips, and molecular identification was done by partial 16S rRNA gene using PCR. The *S. typhi* isolates were categorized into multidrug-resistant (MDR), 13 (13.68%), and extensively drug-resistant (XDR), 82 (86.31%), and their resistance patterns were recorded. Ampicillin (98.94%) and chloramphenicol (93.68%) showed the highest antibiotic resistance profiles, while azithromycin and meropenem exhibited no resistance. Numerous mutations were found in *acrA*, *acrB*, and *tolC* genes after sequencing; TolC (MDR) showed the highest score (16 points), and AcrB (MDR) displayed the lowest score (9 points). I-Mutant 2.0 was used to assess mutations and calculate the reliability index (RI), whereas trRosetta and Discovery Studio were used to predict and refine 3D protein models. Consensus sequences of the selected genes were analyzed to construct phylogenetic trees illustrating evolutionary relationships with other *Salmonella enterica* serovars. The study emphasizes the concerning multidrug resistance of *S. typhi* isolates as well as notable mutations (genetic changes) that may affect efflux pump activity and contribute to resistance.

Introduction

Typhoid fever is a multi-systemic infection caused by *Salmonella enterica serovar* Typhi (*Salmonella typhi*) [1]. It has become a global issue due to growing antibiotic resistance to anti-typhoidal medicines in practice [2]. In addition to making the disease more severe, antibiotic-resistant *Salmonella* infections raise the expense of treatment, necessitate hospitalization, and cause financial losses [3]. *Salmonella typhi* (*S. typhi*) is a facultative anaerobic, rod-shaped, Gram-negative bacterium that belongs to the family Enterobacteriaceae [4]. It is harmful because of its polysaccharide capsule, which shields it from phagocytosis [5]. The oral-fecal route is frequently used to transmit *S. typhi*, which can be lethal without treatment [6]. The most common symptoms include fever, nausea, vomiting, stomach discomfort, and irregular bowel movements [7]. It is one of the most common bacterial diseases that people get worldwide. Typhoid fever is more common among people living in low- and middle- income countries (LMICs) [8]. Insufficient sanitation,

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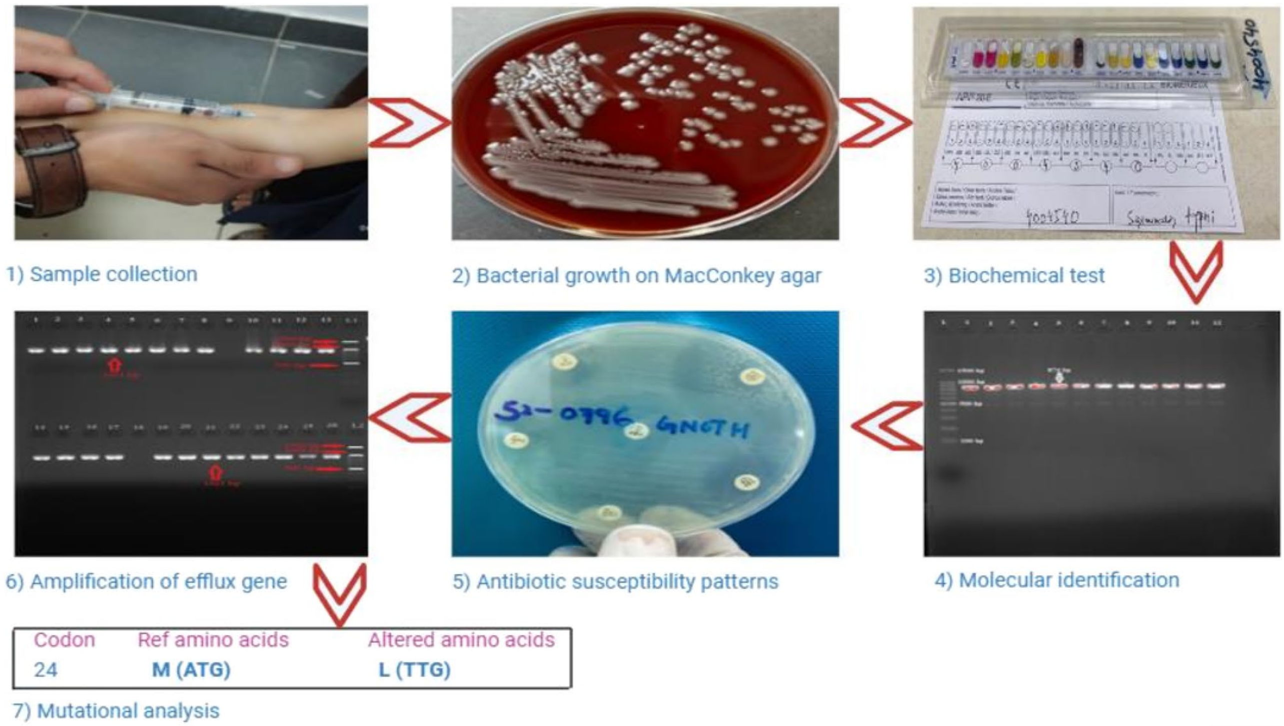
lack of clean water, and insufficient sewage infrastructure are the main causes of rising infection burden in LMICs [9]. It is estimated that 11–21 million individuals globally per year get typhoid fever, which kills 120,000–160,000 people annually [10]. The Global Health Data Exchange reports that about 116,800 deaths in 2017 were attributable to typhoid illness. Over 79,000 mortality occurred in South Asia, with 6,700 of them occurred in Pakistan [11]. Despite the tremendous progress in research and medicine, millions of people are still at risk of fatal enteric fever, which can result in disability and even death [12]. A significant threat to effective treatment is due to the emergence of *S. typhi*-resistant strains, such as multidrug-resistant (MDR) and extensively drug-resistant (XDR) strains [13]. Antibiotics for typhoid fever include trimethoprim/sulfamethoxazole, chloramphenicol, ampicillin, ceftriaxone, ciprofloxacin, azithromycin, tigecycline, and meropenem. Azithromycin resistance has also been reported in a few instances [14]. Antibiotic misuse, such as selling over-the-counter (OTC) medications and prescribing antibiotics without valid medical indications, plays a significant role in the emergence of drug-resistant bacteria [15].

The antibiotic resistance mechanisms for bacteria include antibiotic degradation and enzymatic modification, changing drug targets to modify the affinity of antibiotics, and changing bacterial surfaces by changing the expression of external membrane proteins and active bacterial efflux for medicines [16]. In addition to contributing to antibiotic resistance, efflux pumps in *Salmonella* and other enteric bacteria are involved in the transport of harmful substances like drugs, bile salts, toxins, and environmental chemicals, hence increasing bacterial survival [17]. These pumps in bacteria reduce cellular drug accumulation and are frequently linked to drug resistance [18]. *Salmonella's* emergence of antibiotic resistance has been observed in both humans and animals, raising the possibility of a significant public health concern [19]. Through analysis of bacterial genome sequences, potential drug resistance genes in *Salmonella* can be identified [20]. *Salmonella*, with the exception of the SMR family, encodes at least eleven MDR pumps from each family. The nine functioning drug efflux pumps of them include AcrAB, AcrEF, AcrD, MdsAB, MdtABC, EmrAB, MdtK, MdfA, and MacAB. Based on the recent research, two novel families of *Salmonella* efflux pumps have been identified. These new drug efflux systems are members of the p-aminobenzoyl-glutamate transporter (AbgT) family and the protobacterial antimicrobial compound efflux (PACE) superfamily [21]. According to the sequence similarities, these nine efflux pumps belong to four families: the major facilitator (MF) family (MdfA and EmrAB); the resistance-nodulation-division (RND) family (AcrAB, AcrEF, AcrD, MdsAB, and MdtABC); the multidrug and toxic compound extrusion (MATE) family (MdtK); and the ATP-binding cassette (ABC) family (MacAB) [22]. *Salmonella* has an outer membrane protein channel called the TolC [23]. In *Salmonella,* the tripartite complex of AcrB named AcrAB-TolC has a wide variety of substrates and is well-characterized as an RND pump [24]. This pump, along with other RND pumps, is a major modulator of multidrug resistance in Gram-negative bacteria, including several Enterobacteriaceae [25]. Alarmingly, it has been suggested recently that mutations in RND-type exporters themselves might promote MDR by increasing efflux activity [26]. Drug-binding affinity can be changed by mutation in efflux pump genes that change the substrate-binding pocket shape. The pumps capacity to identify and extrude antibiotics may be improved by these structural alterations, which would reduce intracellular drug concentrations and increase multidrug resistance [27]. Five potential drug efflux pumps from the RND family, for instance, have been annotated [28]. Despite the extensive research on AcrAB, AcrD, and AcrEF, little is known about the roles of other efflux pumps in virulence and drug resistance [29].

Bacterial colonization is facilitated by efflux pump channel-mediated drug resistance. So, it is essential to look into the genetic characterization of *S. typhi* strains in order to monitor efflux pump genes within the population and improve antibiotic therapy. To the best of the author's knowledge, no other studies have been done on the genomic perspective of efflux pumps in drug-resistant *S. typhi* isolates in district Peshawar, Pakistan. As a result, the aim of this study is to find out the genetic characteristics of efflux pump genes and drug resistance patterns of *S. typhi* isolates in said region. This information could be utilized to develop new drugs that could target these genes specifically or even to identify some distinct, specific targets that might be employed as novel therapeutic targets. Additionally, it will facilitate the rapid identification of *S. typhi*-resistant strains and will enhance infection control in an efficient manner.

Materials and Methods

During this study, standard microbiological techniques were used and carried out at the Center of Biotechnology and Microbiology, University of Peshawar, and District Peshawar's tertiary care hospitals, such as Hayatabad Medical Complex (HMC), Lady Reading Hospital (LRH), and Khyber Teaching Hospital (KTH). The clinical samples were collected from the patients admitted to the respective wards of the aforementioned hospitals. Automated blood culture systems were used to process ($n = 2950$) blood samples that were aseptically collected from suspected typhoid patients across a range of age groups and genders and were placed in EDTA tubes. Scheme 1 shows a comprehensive overview of the entire



Scheme 1 Methodology flowchart for the current study

process. Every patient was given an informed permission form on a mandated proforma prior to the collection of blood samples.

Isolation and Identification

The samples were cultured on blood agar (Merck, USA, Rahway, NJ), Salmonella Shigella (SS) agar, and MacConkey agar (USA Merck, Rahway, NJ) and were incubated for 24 h at 37 °C [30]. The isolates were identified biochemically by Analytical Profile Index (API) 20-E strips (Bio-Merieux, France) and Gram staining (Merck, USA, Rahway, NJ) [31].

Molecular-Level Identification

The DNA from bacterial isolates was extracted using Thermo Scientific GeneJET Genomic DNA purification kits in compliance with the manufacturer's protocol in order to identify the *S. typhi* at the molecular level by using partial 16S rRNA gene primers (Table 1) [32]. The extracted DNA and/or amplified 16S rRNA gene products were assessed by 1.5% agarose gel electrophoresis and visualized using a gel documentation system [33].

Antibiotic Susceptibility Patterns

According to Clinical and Laboratory Standards Institute (CLSI) guidelines, antibiotic susceptibility patterns of

Table 1 Primers sequences under optimized conditions

Gene	Primers sequences	Gene size (bp)	Amplification condition	Cycle	References
16S rRNA	F: AGGCCTTCGGGTTGTAAAGT R: TTATGAGGTCCGCTTGCTCT	874	Annealing: 55/30 (°C/s)	35	Designed in this study
acrA	F: TGATGCTCTCAGGCAGCTTA R: TTCAATCCGTCAGTCACCAG	1021	Annealing: 55/30 (°C/s)	35	Designed in this study
acrB	F: GTAGCGCAATATCCGACGAT R: CAGCGATGTTCTGTCGAATG	3056	Annealing: 54/30 (°C/s)	35	Designed in this study
toIC	F: AGAAATTGCTCCCCATCCTT R: GGATTGCCGTTATTGCTGTT	1453	Annealing: 55/30 (°C/s)	35	Designed in this study

identified bacterial isolates were ascertained using Kirby-Bauer disc diffusion methods [34] against specific antibiotic discs. After applying antibiotic discs and inoculating pure cultures of bacterial isolates (0.5 McFarland standards) on sterile Muller Hinton Agar (MHA), the mixture was incubated for 24 h at 37 °C. According to CLSI guidelines [35], inhibition zones were assessed and classified as resistant and sensitive (Table 3).

Efflux Pump Genes Amplification by Polymerase Chain Reaction

The *acrA*, *acrB*, and *tolC* efflux pump genes were amplified by polymerase chain reaction (PCR), using specific primers in accordance with the antibiotic resistance pattern (MDR and XDR). PCR was performed using 2 µL of DNA sample, 11.5 µL of nuclease-free water, 12.5 µL of Taq Master Mix (Thermo Fisher Scientific™, Waltham, MA, USA), and 0.5 µL of forward and reverse primer (from Seoul, Korea's oligonucleotide Microgen). The selected efflux pump genes were amplified under optimum circumstances, and the results were verified by using 1.5% agarose gel electrophoresis and visualized with a gel documentation system (Table 1) [36].

DNA Sequencing and Mutational Analysis

The amplified PCR products of selected genes were sequenced via "23rd Pair Lab Empowering Solutions: Leaders in Molecular Bioinformatics, Metabolomics, and Multi-Omics Big Data, Karachi, Pakistan" following purification with a purification kit (Waltham, Massachusetts, USA: Thermo Scientific™ GeneJET Purification Kit for PCR). The GenBank National Center for Biotechnology Information (NCBI) database was used to obtain FASTA sequences following the sequencing of the aforementioned genes. Basic Local Alignment Search Tool (BLAST) and BioEdit 7.2 software were used to compare the FASTA sequences of the selected genes in order to confirm their prevalence and mutations in bacterial isolates [37].

I-Mutant Prediction

The data were analyzed for non-synonymous mutations, and the reliability index (RI) and pathogenic consequences of the identified mutations were predicted under standard conditions (pH 7, temperature 25 °C) using I-mutant software 2.0 [38].

Prediction and Validation of Protein Model

BioEdit was used to align the consensus and reference amino acid sequences for the efflux pump proteins and examined with trRosetta to forecast the orientations and separations between residues. Gradient-based energy reduction was used to create the structural models, which were then verified with Discovery Studio 25.1.0.0. The models with the highest quality and stability were selected, visualized, and evaluated for prospective uses and structural integrity [39].

Phylogenetic Analysis

Homologous sequences were aligned using MUSCLE or ClustalW and saved in MEGA format for phylogenetic analysis using Molecular Evolutionary Genetic Analysis (MEGA-11). After choosing the optimal substitution model, a phylogenetic tree was created using "Unweighted Pair Group Method with Arithmetic-means (UPGMA) and Maximum Parsimony (MP)" techniques. After evaluating the tree's robustness using bootstrap replicates, the refined tree was saved in the desired format, annotated, and displayed for additional study [40].

Statistical Analysis

Descriptive statistics were calculated for the current study, including mean, median, mode, standard deviation, variance, range, minimum, and maximum (Table S1). An examination of the gender frequency distribution was also conducted (Table S2). A chi-square test via $P \leq 0.05$ was conducted using SPSS version 27.0.1, and $n-1$ degrees of freedom were conducted to study the correlation among the patient's gender and resistance to different antibiotics (Tables S3–S18).

Nucleotide Sequence Accession

The partial 16S rRNA gene sequence of *S. typhi* for both MDR and XDR isolates has been deposited in the National Center for Biotechnology Information (NCBI) GenBank database under the accession nos. PV945086 and PV945087, respectively.

Results

During the present study, blood samples ($n = 2950$) were collected from suspected typhoid-infected patients. Of these, 380 (12.88%) were found positive for the bacterial growth, including 144 (37.89%) identified as Gram-positive and 236 (62.10%) as Gram-negative. Out of Gram-negative bacteria, 95 (25%) patients had *S. typhi*, while the remaining 141 (37.10%) had other types of Gram-negative bacteria. The *S. typhi* patients include 63 (66.31%) males and 32 (33.68%) females. Age group 0–10 years had the highest *S. typhi* ratio, at 60 (63.15%), followed by

11–20 years, at 21 (22.10%), and those over 60 years had the lowest ratio, at 1 (1.05%) (Table 2).

Identification of Isolates

To identify the isolates, they were cultured on Salmonella Shigella (SS), Blood, and MacConkey agar. Gram staining was done under a microscope (pink colonies), and then biochemical tests were performed using API 20-E strips. Finally, molecular identification by partial 16S rRNA gene was performed (Fig. S1).

Patterns of Antibiotic Susceptibility in Clinical Isolates

Patterns of antibiotic sensitivity to specific antibiotics, such as ampicillin (AMP), sulfamethoxazole-trimethoprim (SXT), meropenem (MEM), ciprofloxacin (CIP), chloramphenicol (C), cefotaxime (CTX), ceftazidime (CAZ), and meropenem (MEM), were documented. Resistance to AMP (98.94%) was the most common, followed by C (93.68%), while no resistance against AZM and MEM was recorded (Table 3).

Table 2 The *Salmonella typhi* isolates frequency and percentage distribution by age groups

Parameters	*S. typhi* (*n* = 95)
Frequency (%)	
Gender	
Male	63 (66.31)
Female	32 (33.68)
Age groups	
00 to 10	60 (63.15)
11 to 20	21 (22.10)
21 to 40	11 (11.57)
41 to 60	2 (2.10)
Above 60	1 (1.05)

Genotypic Identification of Efflux Pump Gene Producers

The *acrA*, *acrB*, and *toIC* genes were genotypically screened in all 95 *S. typhi* isolates (Fig. S2). Out of all 95 isolates of *S. typhi*, 89 (93.68%) contained *acrA*, 91 (95.78%) contained *acrB*, while all 95 (100%) isolates contained the *toIC* gene (Table 4).

Mutational Analysis of acrA, acrB, and tolC Efflux Pump Genes

The sequencing data was analyzed by various bioinformatics tools, including NCBI, BLAST, and BioEdit. Multiple nucleotide mutations at different points were detected in selected genes of *S. typhi* (MDR and XDR) isolates. The *acrA*, in XDR isolate include mutations like C → G at position 28 and C → A at 984, while MDR isolate had mutations like A → T at position 70, G → T at position 102, and an insertion of T at 787. Key alterations such as G → A at 22, A → G at 2216, and G → C at 2680 were also observed in *acrB* in MDR isolate, whereas A → C at 310 and G → A at 939 were observed in XDR isolate. A → C at 66, G → T at 501, and T → C at 1361 were *toIC* mutations found in MDR isolate, while C → G at 32, T → A at 842, and G → T at 1359 were found in XDR isolate (Table 5).

I-Mutant Predictions

As predicted by I-mutant, the mutation predictions and RI were determined under standard conditions (pH 7, temperature 25 °C). The total identified mutations in the AcrA protein of MDR isolates, including M → L at position 24, F → S at 99, and V → A at 196 (RI = 9), were expected to reduce stability and perhaps affect AcrA function in antibiotic

Table 4 Distribution of efflux pump genes in clinical isolates

Gene	Positive	Negative
acrA	89 (93.68%)	6 (6.31%)
acrB	91 (95.78%)	4 (4.21%)
toIC	95 (100%)	0 (0.00%)

Table 3 Antibiotic susceptibility patterns of *Salmonella typhi* isolates against specific antibiotics

Antibiotics	*Salmonella typhi* (*n* = 95)		Antibiotics	*Salmonella typhi* (*n* = 95)	
	Sensitive (%)	Resistant (%)		Sensitive (%)	Resistant (%)
Chloramphenicol (C)	6 (6.31)	89 (93.68)	Cefotaxime (CTX)	12 (12.68)	83 (87.36)
Ceftazidime (CAZ)	24 (25.26)	71 (74.73)	Ciprofloxacin (CIP)	17 (17.89)	78 (82.10)
Meropenem (MEM)	95 (100)	0 (0.00)	Sulfamethoxazole-trimethoprim (SXT)	24 (25.26)	71 (74.73)
Azithromycin (AZM)	95 (100)	0 (0.00)	Ampicillin (AMP)	1 (1.05)	94 (98.94)

Table 5 Mutations detected in selected efflux pump genes

Gene	Isolate types	Position	Mutation	Position	Mutation	Position	Mutation
acrA	MDR	70	T replaced A	102	T replaced G	172	C replaced A
		262	T replaced C	296	C replaced T	430	T replaced C
		523	G replaced A	587	C replaced T	628	C replaced T
		730	G replaced A	787	Insertion of T	888	T replaced C
		971	G replaced A				
acrA	XDR	28	G replaced C	62	A replaced G	84	G replaced A
		124	A replaced G	167	C replaced G	282	C replaced G
		326	C replaced G	490	A replaced C	500	T replaced A
		984	A replaced C				
acrB	MDR	22	A replaced G	165	A replaced G	178	G replaced A
		567	A replaced T	690	C replaced T	984	C replaced G
		2216	G replaced A	2328	G replaced C	2680	C replaced G
acrB	XDR	310	C replaced A	644	A replaced T	784	G replaced C
		939	A replaced G	996	A replaced C	2013	C replaced G
		2286	C replaced T	2572	T replaced G	2801	A replaced T
		2905	G replaced C	3015	C replaced A		
toIC	MDR	66	C replaced A	164	A replaced G	305	C replaced G
		501	T replaced G	622	T replaced G	742	C replaced G
		794	T replaced C	881	A replaced T	902	T replaced A
		938	C replaced T	991	T replaced G	1094	A replaced G
		1124	A replaced C	1361	C replaced T	1382	C replaced T
		1393	C replaced A				
toIC	XDR	32	G replaced C	164	A replaced G	485	T replaced G
		630	C replaced T	699	C replaced T	794	T replaced C
		842	A replaced T	938	C replaced T	1094	A replaced G
		1124	A replaced C	1146	A replaced G	1359	T replaced G

resistance. G → D at position 21 (RI = 1), R → P at position 56 (RI = 5), and D → V at position 167 (RI = 7) were among the most disruptive mutations found in XDR isolates. Nevertheless, there was just one stabilizing mutation found, Q → E at position 10 (RI = 6). In the AcrB protein, mutations like T → A at position 60 (RI = 8) and A → T at position 8 (RI = 7) were anticipated to reduce protein stability in MDR isolates, perhaps affecting efflux pump function. Destabilizing substitutions were also seen in XDR isolates, such as S → R at position 104 (RI = 3), A → S at 858 (RI = 9), and L → Q at 934 (RI = 9), which suggested substantial structural alterations. Nonetheless, it was projected that two mutations, K → N at 1005 (RI = 0) and Q → E at 969 (RI = 1), would improve stability, albeit with poor dependability. In the ToIC protein, every mutation found was expected to reduce protein stability, which could have an effect on efflux pump performance. Several high-reliability substitutions, including A → S at 331, R → Q at 365, and P → Q at 375 (all RI = 9), were found in MDR isolates, along with critical disruptive alterations R → H at position 55 (RI = 9), W → C at 167 (RI = 7), and A → S at 208 (RI = 8). Strongly disruptive mutations, such as R → H at position 55 (RI = 9), R → L at 162 (RI = 9), and Q → H at 453 (RI = 8), were also seen in XDR isolates (Fig. S3). The efflux pump mechanism may be impacted by these structural changes in the selected genes, which could lead to antibiotic resistance in both MDR and XDR strains.

Prediction and Validation of Protein Structural Models

Using the trRosetta, 3D models of the target protein upon consensus and reference sequence for selected genes were predicted and verified by Discovery Studio. Rosetta energy ratings, contact probability analysis, and Ramachandran plot evaluations were used to evaluate stability. The model's structural integrity was further validated, proving that trRosetta and Discovery Studio work together to provide a trustworthy and realistic structural model.

Estimated Template Modeling Score (TM-Score)

Variations in structural similarity between reference and consensus/mutated sequences were found in MDR and XDR isolates using the estimated Tm-score analysis for selected genes. From reference (0.315 in MDR and 0.305

in XDR) to consensus (0.405 in MDR and 0.403 in XDR), AcrA showed a rise in Tm-score, indicating possible structural stability brought on by mutations. Despite resistance type, AcrB maintained high Tm-scores with little fluctuation (0.963–0.966 in MDR and 0.958–0.957 in XDR), suggesting a structurally conserved efflux pump. On the other hand, ToIC consistently displayed low Tm-scores (0.100 for both reference and consensus sequences in MDR and XDR), suggesting a structure that is either extremely flexible or intrinsically unstable. These results demonstrate that ToIC shows possible structural instability, AcrA experiences structural alterations, and AcrB stays stable, all of which may have an impact on their involvement in antibiotic resistance. The most reliable prediction was indicated by the AcrB-MDR model for consensus sequence of amino acids, which had the highest TM-score (0.966). Four models, including those predicted by the reference and consensus amino acid sequences of the ToIC (MDR and XDR), had the lowest TM-score (0.100), indicating less reliable prediction (Fig. 1).

Predicted Per-Residue Local Distance Difference Test (pLDDT)

Protein structure confidence is measured by the projected pLDDT scores (0–100), where values above 70 indicate well-defined regions and below 50 indicate low-confidence sections. In this study, high confidence in core areas and low confidence in flexible loops were found using pLDDT analysis, which shed light on the predictability of the model (Fig. 2).

3D Protein Structural Models

Red represents high confidence, and blue indicates low confidence in the 3D protein structural models produced by trRosetta, which are shown as ribbon models that emphasize important structural components. These models offer insights into protein folding and domain organization and serve as a useful framework for structural comprehension. They are based on deep learning and sequence co-evolutionary data (Fig. 3).

Models Quality Assessment by Discovery Studio

The model quality assessment ensures that protein-predicted structural models are reliable and accurate. Ramachandran plot analysis verifies the correct backbone conformation. The TM-score evaluates accuracy by comparing models. Energy scoring and contact probability analyses are used to further assess stability. Together, these analyses validate the anticipated model's structural soundness and conformity to known protein folds.

Ramachandran Plot for Predicted Protein Models

With the majority of residues in authorized and preferred regions, the Ramachandran plot analysis demonstrated good structural correctness and validated the stereochemical quality of the predicted protein models. Minor adjustments were suggested by a few residues in less desirable regions. All things considered, the investigation confirmed the models' dependability for additional structural and functional research (Fig. 4).

Phylogenetic Analysis

The *acrA* genes show the genetic diversity and evolutionary relationships between *Salmonella enterica* serovars, with a focus on strains associated with MDR and XDR isolates. While more distant strains like AUSMDU and PNUSAS

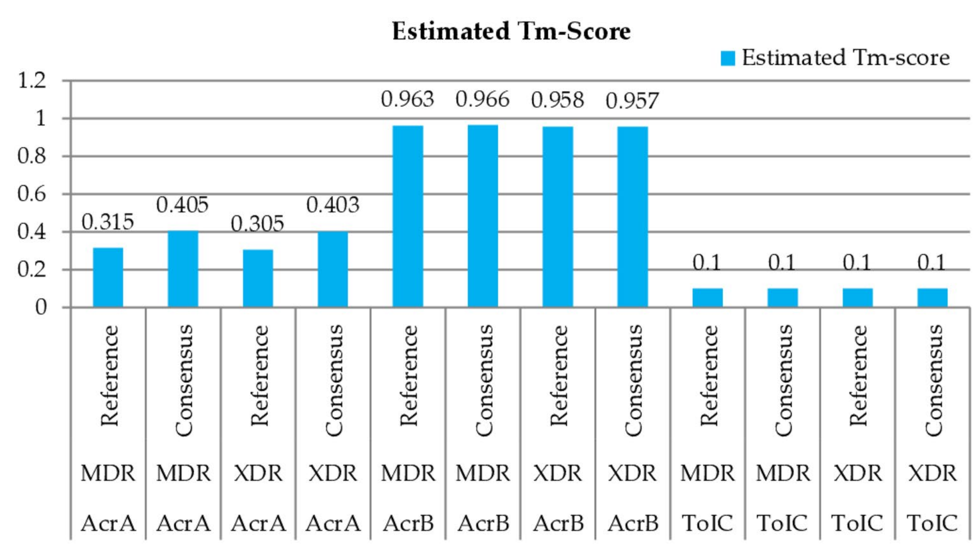
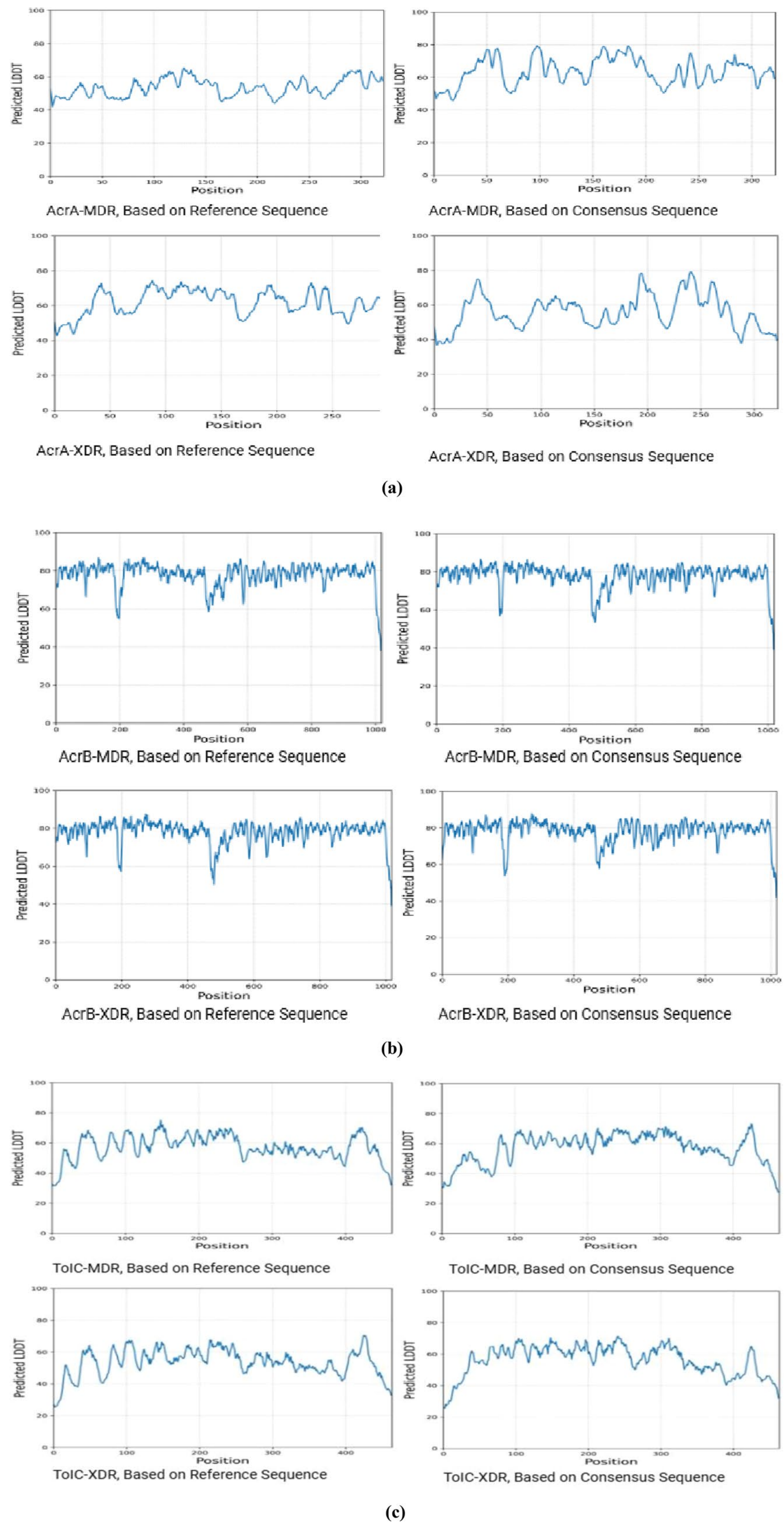


Fig. 1 Tm-score estimation for the predicted structural models by trRosetta

Fig. 2 The models' pLDDT for MDR and XDR isolates of protein. **a** AcrA, **b**, AcrB, and **c** TolC



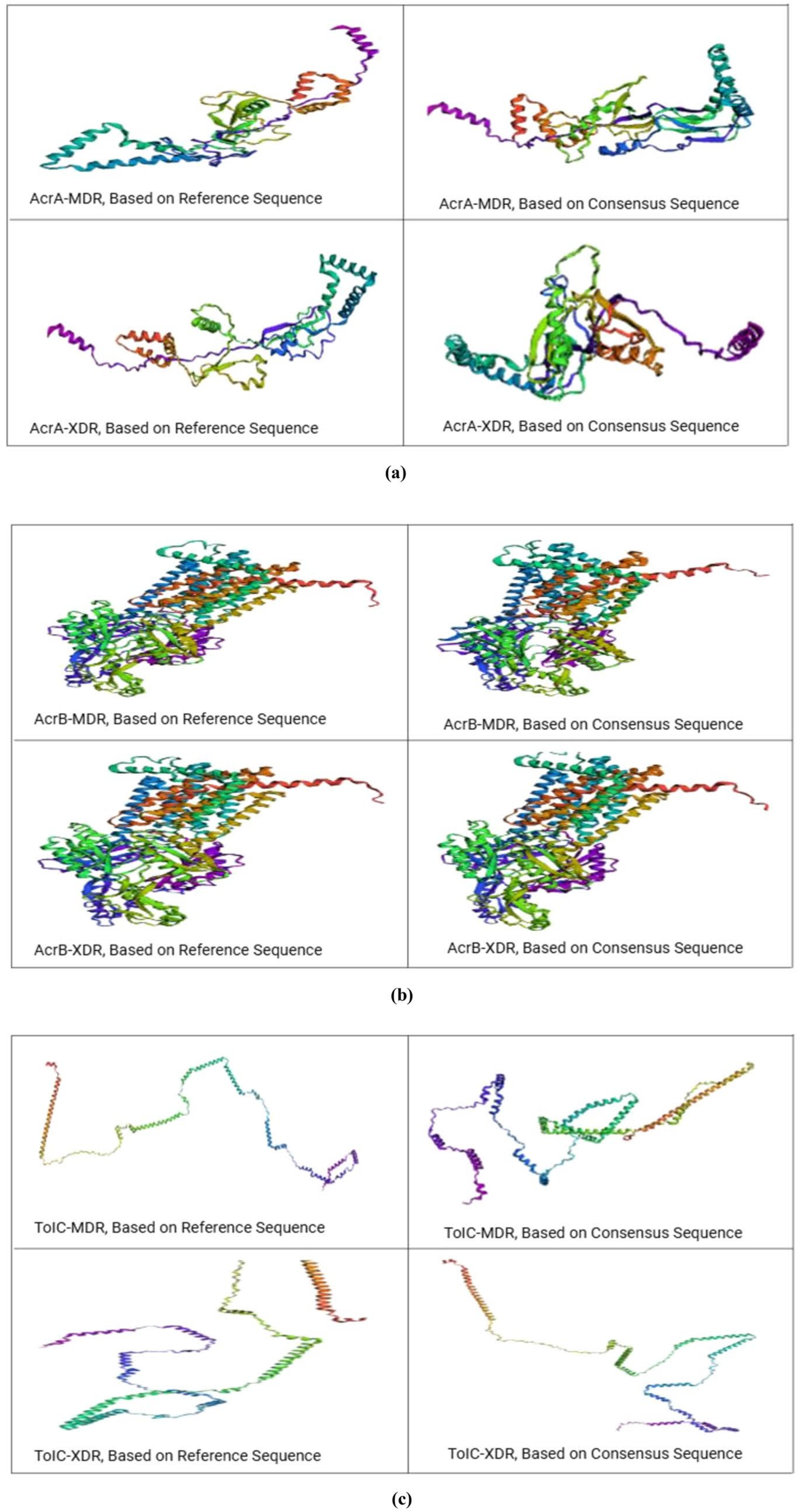
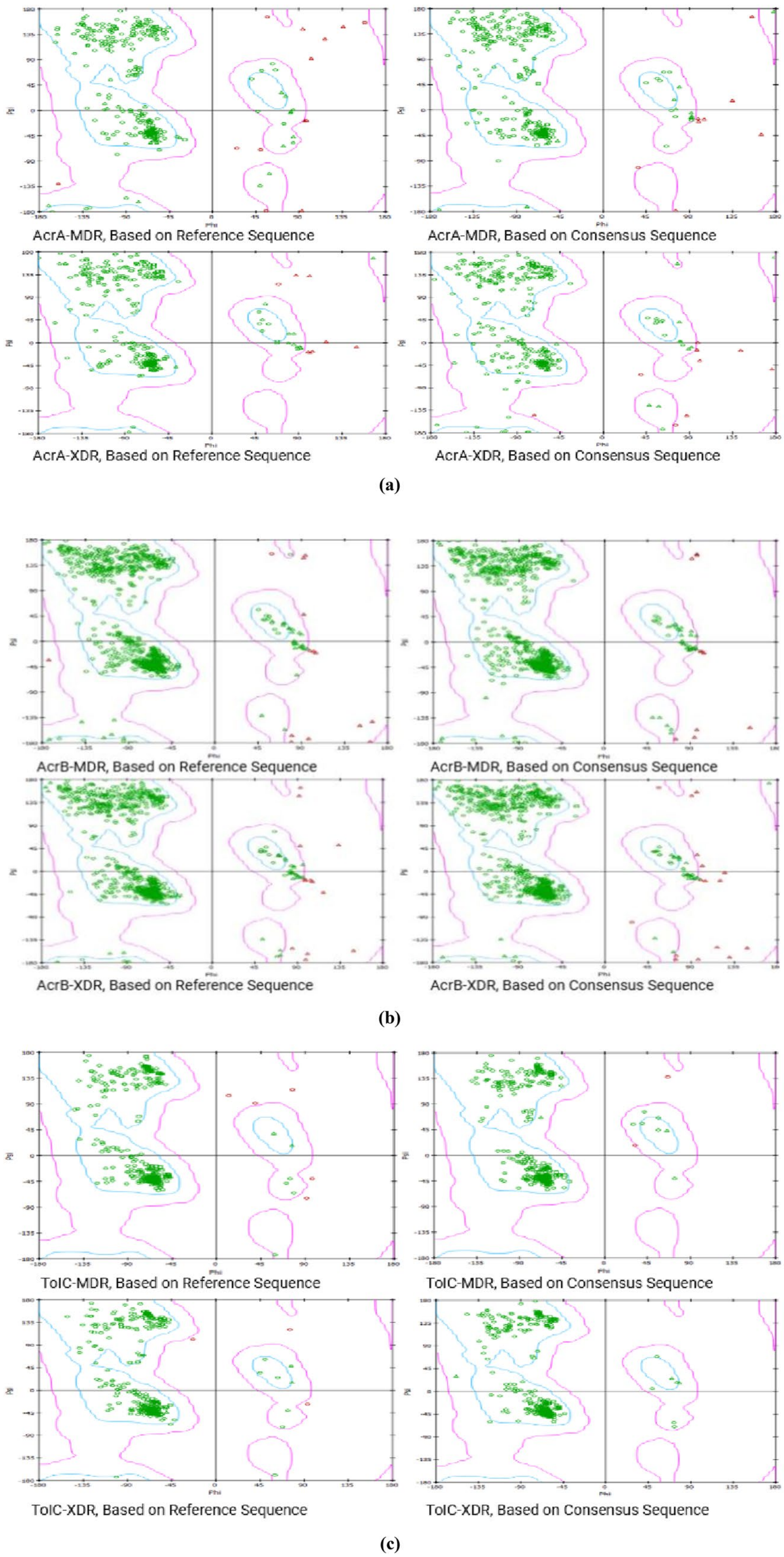


Fig. 3 The 3D structural models predicted for MDR and XDR isolates of protein. **a** AcrA, **b** AcrB, and **c** TolC

Fig. 4 The Ramachandran plot for MDR and XDR isolates of genes. **a** AcrA**, b** AcrB, and **c** TolC



highlight greater diversity, clustering of strains like 311189 and 343077 of *S. typhi* indicates shared genetic lineages. In terms of *acrB* genes, the figures contrast several *Salmonella enterica* serovars, including Dessau and Senftenberg, emphasizing the existence of the AcrB (MDR) and AcrB (XDR) genes, which are essential to our study. The importance of strains such as CP047424.1 and CP029038.1 in resistance research is highlighted by their recurrence. A genomic comparison of several *Salmonella enterica* serovars, including Typhi and Typhimurium, is shown in Fig. 5c for *tolC* genes. Of particular interest to our study are the *tolC* (MDR) and *tolC* (XDR) genes. The significance of examining resistance mechanisms is highlighted by the inclusion of different strains, including CP029873.1 and CP156169.1. Given that they were found in several strains, they may play important roles in MDR and XDR. Their distribution and effects in fighting drug-resistant *Salmonella* infections are best understood with the use of this genetic data (Fig. 5).

Discussion

Antibiotic resistance is a major public health concern that raises death and morbidity rates worldwide. The antibiotic resistance mechanisms for bacteria include antibiotic degradation and enzymatic modification, changing drug targets to modify the affinity of antibiotics, and changing bacterial surfaces by changing the expression of external membrane proteins and active bacterial efflux for medicines [16]. Mutation in efflux pump genes that alter the geometry of the substrate-binding pocket can alter the drug-binding affinity. These structural changes may enhance the pumps ability to recognize and expel antibiotics, lowering intracellular drug concentrations and boosting multidrug resistance [27]. Antibiotics are therefore ineffective against bacteria and make the treatment of bacterial infections more difficult. The results of the current study confirmed this phenomenon, which affects both public health and the economy. Typhoid-suspected blood samples were processed during the current investigation ($n = 2950$), and 380 (12.88%) of the samples were found positive for the bacterial growth, whereas 95 (25%) of the 380 patients tested positive for *S. typhi* infection. Additionally, it was observed that 32 (33.68%) of the 95 *S. typhi*-positive samples were female, while 63 (66.32%) were male. Our findings are consistent with those of Kim et al., who discovered that 31.11% of the females and 68.89% of the males had *S. typhi* infections [41]. Zakir et al. also reported that *S. typhi* infections were discovered in more males (60.67%) than females (39.33%) [42]. Out of 511 blood samples, 47 (9.19%) contained *S. typhi*, with 33 (70.23%) coming from male patients and 14 (29.77%) from female, as reported in another study [43]. Changes in the sample population (demographics, health status, and behavior), time, environmental factors (water quality, sanitation, and public health infrastructure), and detection methods (laboratory techniques, diagnostic criteria, and blood sample quality) may all contribute to the gender-based variations in *S. typhi* prevalence. According to these studies, males are more likely than females to get *S. typhi*. Male infections may be more common because they are more vigorous in environments where typhoid is prevalent. Additionally, men might be more likely than women to be tested or seek medical attention, which can lead to a privileged detection rate. Contrary to previous research, the current study indicated that maximum ratio of the *S. typhi* was 60 (63.15%) in age group 0–10 years; that may be the result of numerous contacts or compromised immune systems. This was followed by 21 (22.10%) in age group 11–20 years, consistent with the literature [42].

During the present study, *S. typhi* showed resistance to C, CAZ, CTX, CIP, SXT, and AMP and sensitivity to AZM and MEM. The antibiotics AZM and MEM are recommended for the treatment of *S. typhi.* These finding are in agreement with the literature [44]. In the present research, 82 (86.31%) out of 95 *S. typhi* isolates belong to XDR, and 13 (13.68%) belong to non-XDR. The large number of XDR isolates in the current research may be caused by genetic variables, abuse or misuse of antibiotics, spread of resistance strains, poor infection control procedures, and/or geographic factors. Our findings are consistent with the study by Fatima et al., which discovered 34% non-XDR isolates and 66% XDR isolates [45]. In the current study, *acrA*, *acrB*, and *tolC* genes were found in 93.68%, 95.78%, and 100%, respectively, in *S. typhi* isolates. These results are in line with earlier research that found the AcrAB-TolC efflux system was very prevalent in clinical strains of *S. typhi,* indicating that it plays a conserved role in both intrinsic and acquired multidrug resistance [46]. In another study it was found that these genes were present in *Salmonella* spp. at a frequency of over 90%, demonstrating their extensive distribution and crucial role in antibiotic efflux [47, 48]. Our isolates' somewhat lower prevalence of *acrA* and *acrB* than *tolC* might be due to strain-specific gene loss, natural sequence variability, or regulatory changes that impact PCR detectability. The *tolC's* multifunctional involvement in drug resistance, toxin production, and outer membrane integrity is further supported by its universal presence. All things considered, our research highlights the potential of the AcrAB-TolC system as a target for innovative therapeutic approaches and supports the notion that it plays a significant role in *S. typhi's* development of antibiotic resistance. In current study, the selected efflux pump gene sequences of *S. typhi* isolates were analyzed by different bioinformatics tools, including NCBI, BLAST, BioEdit 7.2, and MEGA-11. Prior research conducted by Hassan et al. also utilized these tools for DNA sequence analysis [49], and Ziaee et al. also used similar

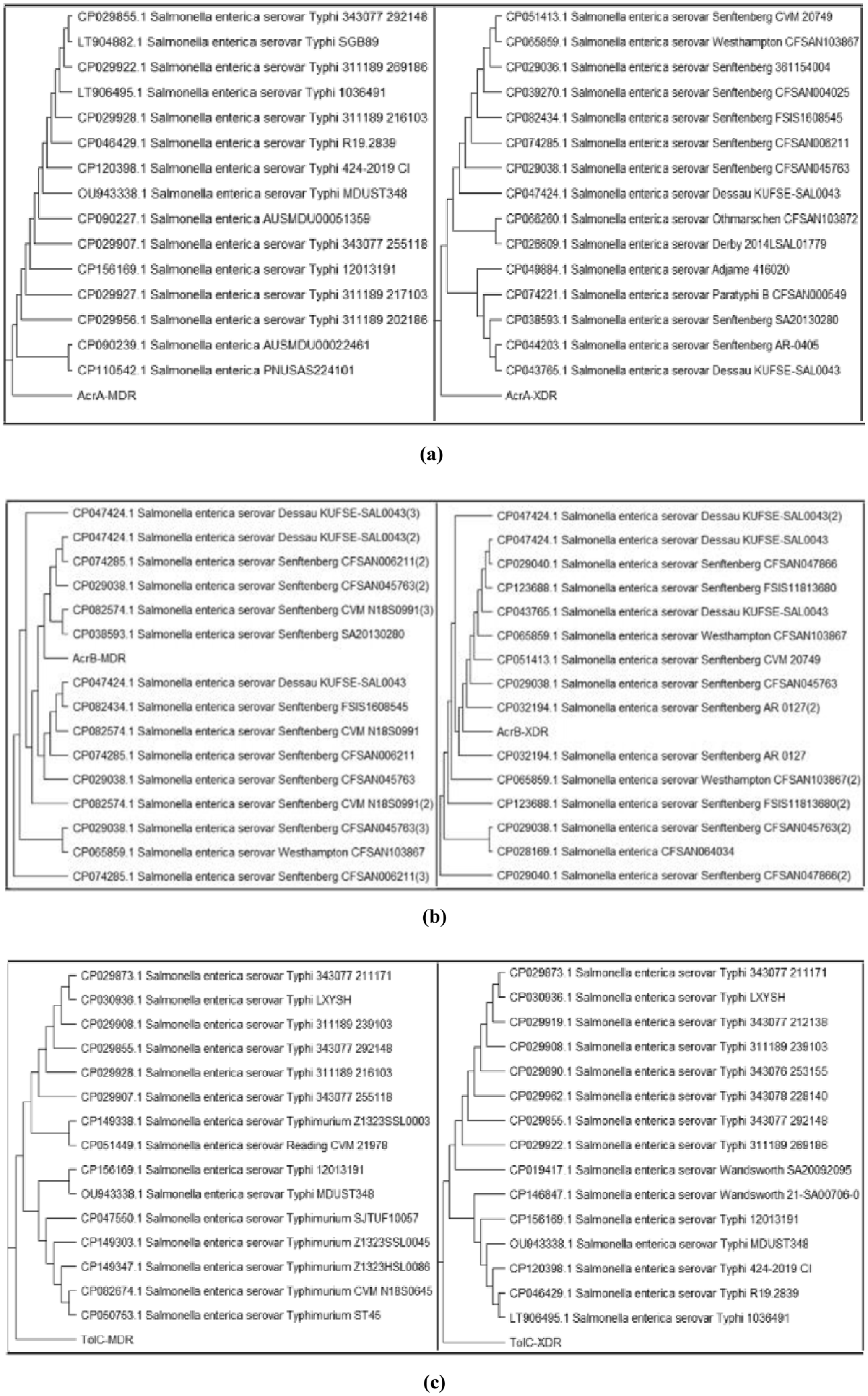


Fig. 5 Phylogenetic tree for *MDR and XDR isolates of genes.* **a** *acrA* by UPGMA. **b** *acrB* by MP and **c** *tolC* by UPGMA using MEGA-11

tools for nucleotide mutation analysis and MEGA-6 for phylogenetic analysis [50]. Amino acid sequence mutations during transcription are caused by DNA sequence mutations. The reference and consensus sequences of amino acids for the selected genes were obtained by BioEdit. Under standard conditions (pH 7, 25 °C), I-Mutant 2.0 was used to predict changes in protein stability caused by point mutations. I-Mutant 2.0 was also used in similar studies by Masso & Vaisman and Capriotti et al. to predict changes in protein stability [51, 52]. According to our computational analysis using I-Mutant, the majority of mutations found in the *acrA, acrB*, and *tolC* genes of MDR and XDR *S. typhi* isolates are predicted to destabilize protein structures, especially at functionally significant domains, which may impair efflux pump performance and contribute to antibiotic resistance. These results are consistent with earlier research, which stated that mutations in efflux pump components can change protein stability, interfere with drug-binding pockets, and hinder pump assembly, all of which can affect resistance phenotypes [27, 53]. Interestingly, despite the detection of a few stabilizing mutations, their low reliability score indicates little functional advantage. Our results are supported by another study, which found that even minor changes in AcrB or TolC protein can have a major impact on substrate recognition or efflux efficiency. This suggests that structural integrity plays a crucial role in multidrug resistance [53]. The trRosetta was used to create the predicted protein structures based on consensus and reference amino acid sequences. Discovery Studio was then used for reliability analysis and validation. Daoqun and Zhang and Radhakumar et al. used similar approaches in their earlier research, using Discovery Studio for model validation and trRosetta for protein modeling [54, 55]. Although our analysis finds several mutations in the *acrA*, *acrB*, and *toIC* efflux pump genes, more research is necessary to fully understand their functional implications. According to structural modeling, certain mutations might have an impact on the efflux or drug-binding efficiency, which could lead to resistance. In terms of therapy, being aware of these modifications might help optimize treatment plans and potentially steer clear of medications that are prone to efflux. However, the small sample size and in silico nature of this study limit its applicability. These results require more biochemical and clinical validation. The present study stated that the models constructed by consensus sequences of AcrA, AcrB, and TolC protein have better structural features than those constructed by reference sequences. These features include more compact folding, improved secondary structure integrity, and better transmembrane domain alignment. These findings align with earlier research indicating that protein structure prediction accuracy and stability are enhanced by consensus or deep learning-guided modeling [56, 57]. Specifically, the observed coherent domain organization in AcrA and TolC and symmetrical folding in AcrB mimic the structural stability improvements documented in prior membrane protein modeling studies. The biological significance of our structural predictions was further supported by prior research, which emphasized the significance of domain interaction and conformational integrity in efflux pump performance [53]. One more study examined the significance of precise structural insights in overcoming multidrug resistance; our results not only confirm the validity of consensus-based modeling for efflux pumps but also serve as a basis for structure-guided inhibitor design against S. typhi [46]. *S. enterica* has other efflux pumps that could also be involved in resistance, even though the AcrAB-TolC system was the main focus of our investigation. The fact that TolC is present everywhere probably indicates that it is a component that is shared by many different systems. As a result, resistance cannot be exclusively ascribed to AcrAB-TolC; for a more thorough understanding, future research should look at other efflux pumps.

Conclusion

This work emphasizes the relevance of intrinsic mechanisms and the necessity of additional functional and therapeutic research by highlighting structural alterations in the *S. typhi* native AcrAB-TolC efflux pump that may impact protein stability and contribute to antibiotic resistance. The frequency of *acrA*, *acrB*, and *toIC* efflux pump genes in *S. typhi* isolates that were MDR and XDR was investigated. These genes were detected in 93.68%, 95.78%, and 100% of the isolates, respectively. During mutational analysis, different levels of mutations were identified and recorded by BioEdit in each of these genes along with their mutation predictions by I-Mutant. The trRosetta was used for protein modeling, and Discovery Studio was used for model validation. This study highlights the significance of antibiotic resistance caused by *S. typhi* efflux pump genes, which aids in the global effort to address the growing antimicrobial resistance (AMR) epidemic. The genomic information generated by this work may prove valuable for future molecular epidemiological studies of *S. typhi* clinical isolates.

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Author Contributions S.A. conceived, overseen and assisted with preparation of manuscript. M. completed the lab work, curated the data and helped with the manuscript's production. N.R. contributed to the project's design and manuscript review. I.K. and A.A. contributed to the work's experimental design. S.A., AAKK and A.S.A. contributed to the Writing and Final manuscript review process. After reading the published version of the manuscript, all writers have given their approval.

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Data Availability To the best of the authors' knowledge, all the data created or analyzed for the study are included in the publication.

Declarations

Conflict of interest There is no conflict of interest disclosed by the authors.

Ethical Approval The Institution Research and Ethical Review Board (IREB) of Khyber Medical College in Peshawar, Pakistan, authorized the study (Document No. 124/ADR/KMC).

Informed Consent Before sampling, informed consent was sought from each patient using prescribed proforma. Every single participant in the study gave their informed consent.

Research Involving Human and Animal Participants Human bloods of suspected typhoid fever were used in this study.

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References

1. Saha T, Arisoyin AE, Bollu B (2023) Diagnostic challenges and the importance of early intervention. Cureus 15:e41831
2. Saleem N, Shahid T (2023) Search for natural remedies for the treatment of *Salmonella typhi* infections. MIC 2:121–134
3. Katiyar A, Sharma P, Dahiya S, Singh H, Kapil A, Kaur P (2020) Genomic profiling of antimicrobial resistance genes in clinical isolates of *Salmonella typhi* from patients infected with Typhoid fever in India. Sci Rep 10:8299
4. Ullah S, Mubarik I, Khan A, Ramzan F, Malik MIU, Ullah S (2024) Antibiogram study and resistance mechanism of *Salmonella typhi* isolated from clinical samples at district DI khan. J Med Health Sci 1:17–40
5. Jia LJ, González K, Orasch T, Schmidt F, Brakhage AA (2024) Manipulation of host phagocytosis by fungal pathogens and therapeutic opportunities. Nat Microbiol 9:2216–2231
6. Salama Y, Chennaoui M (2024) Microbial spoilage organisms in seafood products: pathogens and quality control. Eur J Microbiol Infect Dis 1:66–89
7. Porcari S, Ingrosso MR, Maida M, Eusebi LH, Black C, Gasbarrini A, Ianiro G (2024) Prevalence of irritable bowel syndrome and functional dyspepsia after acute gastroenteritis: systematic review and meta-analysis. Gut 73:1431–1440
8. Mina SA, Hasan MZ, Hossain Z, Barua A, Mirjada MR, Chowdhury MA (2023) The prevalence of multi-drug resistant *Salmonella typhi* isolated from blood sample. Microbiol Insights 16:1–8
9. Shaikh OA, Asghar Z, Aftab RM, Amin S, Shaikh G, Nashwan AJ (2023) Antimicrobial resistant strains of *Salmonella typhi*: The role of illicit antibiotics sales, misuse, and self-medication practices in Pakistan. J Infect Public Health 16:1591–1597
10. Ishtiaq A, Khalil S, Khalil S, Ahmed F, Ahmad B, Ghaffar A, Khalid M, Azam SM, Imran M (2023) Prevalence of typhoid fever among different sociodemographic groups in District Bahawalnagar, Pakistan: prevalence of typhoid fever. Pak J Health Sci 4:138–143
11. Mumtaz MN, Irfan M, Siraj S, Khan A, Khan H, Imran M, Khan IA, Khan A (2024) Whole-genome sequencing of extensively drug-resistant *Salmonella enterica* serovar *Typhi* clinical isolates from the Peshawar region of Pakistan. J Infect Public Health 17:271–282
12. Ehuwa O, Jaiswal AK, Jaiswal S (2021) *Salmonella*, food safety and food handling practices. Foods 10:907–923
13. Khan M, Shamim S (2022) Understanding the mechanism of antimicrobial resistance and pathogenesis of *Salmonella enterica serovar* Typhi. Microorganisms 10:2006–2015
14. Dyson ZA, Klemm EJ, Palmer S, Dougan G (2019) Antibiotic resistance and typhoid. Clin Infect Dis 2:65–70
15. Kotwani A, Bhanot A, Singal GL, Gandra S (2022) Marketing and distribution system foster misuse of antibiotics in the community: insights from drugs wholesalers in India. Antibiotics 11:95
16. Gauba A, Rahman KM (2023) Evaluation of antibiotic resistance mechanisms in gram-negative bacteria. Antibiotics 12:1590–1619
17. Krzyżek P (2024) *Helicobacter pylori* efflux pumps: a double-edged sword in antibiotic resistance and biofilm formation. Int J Mol Sci 25:12222–12236
18. Yamasaki S, Zwama M, Yoneda T, Nishino MH, Nishino K (2023) Drug resistance and physiological roles of RND multidrug efflux pumps in *Salmonella enterica, Escherichia coli* and *Pseudomonas aeruginosa*. Microbiology 169:1–20
19. Lamichhane B, Mawad AM, Saleh M, Kelley WG, Harrington PJ, Lovestad CW, Helmy YA (2024) Salmonellosis: an overview of epidemiology, pathogenesis, and innovative approaches to mitigate the antimicrobial resistant infections. Antibiotics 13:76
20. Wu S, Hulme JP (2021) Recent advances in the detection of antibiotic and multi-drug resistant salmonella: an update. Int J Mol Sci 22:3499
21. Punchihewage-Don AJ, Ranaweera PN, Parveen S (2024) Defense mechanisms of *Salmonella* against antibiotics: a review. Front Antibiot 3:1448796
22. Horiyama T, Yamaguchi A, Nishino K (2010) TolC dependency of multidrug efflux systems in *Salmonella enterica* serovar Typhimurium. J Antimicrob Chemother 65:1372–1376
23. Whittle EE, Orababa O, Osgerby A, Siasat P, Element SJ, Blair JMA, Overton TW (2024) Efflux pumps mediate changes to fundamental bacterial physiology via membrane potential. Antimicrob Chemother 15:1–18
24. Zack KM, Sorenson T, Joshi SG (2024) Types and mechanisms of efflux pump systems and the potential of efflux pump inhibitors in the restoration of antimicrobial susceptibility, with a special reference to *Acinetobacter baumannii*. Pathogens 13:197
25. Eichenberger EM, Thaden JT (2019) Epidemiology and mechanisms of resistance of extensively drug resistant gram-negative bacteria. Antibiotics 8:37
26. Lorusso AB, Carrara JA, Barroso CDN, Tuon FF, Faoro H (2022) Role of efflux pumps on antimicrobial resistance in *Pseudomonas aeruginosa*. Int J Mol Sci 23:15779
27. Nishino K, Yamasaki S, Nakashima R, Zwama M, Hayashi-Nishino M (2021) Function and inhibitory mechanisms of multidrug efflux pumps. Front Microbiol 12:737288
28. Scoffone VC, Trespidi G, Barbieri G, Irudal S, Perrin E, Buroni S (2021) Role of RND efflux pumps in drug resistance of cystic fibrosis pathogens. Antibiotics 10:863
29. Soparkar K, Kinana AD, Weeks JW, Morrison KD, Nikaido H, Misra R (2015) Reversal of the drug binding pocket defects of the AcrB multidrug efflux pump protein of *Escherichia coli*. J Bacteriol 197:3255–3264
30. Khafaji AN (2022) Occurrences and virulence genes detection of *Salmonella typhi* isolates among patients with typhoid fever. Res sq 3:1–13
31. Olubisose ET, Ajayi A, Adeleye AI, Smith SI (2021) Molecular and phenotypic characterization of efflux pump and biofilm in

multi-drug resistant non-typhoidal *Salmonella serovars* isolated from food animals and handlers in Lagos Nigeria. One Health Outlook 3:1–8
32. Scientific TF (2014) Thermo Scientific GeneJET Genomic DNA Purification Kit. User guide.[Internet]. Avalable in: https://www.tools.thermofisher.com/content/sfs/manuals/MAN0012663_GeneJET_-Genomic_DNA_Purification_Kit_UG.pdf
33. Bogiel T, Mikucka A, Kanarek P (2022) Agarose gel electrophoresis-based RAPD-PCR—an optimization of the conditions to rapidly detect similarity of the alert pathogens for the purpose of epidemiological studies. Gels 8:760
34. Bauer RJ, Zhang L, Foxman B, Siitonen A, Jantunen ME, Saxen H, Marrs CF (2002) Molecular epidemiology of 3 putative virulence genes for *Escherichia coli* urinary tract infection–usp, iha, and Iron (*E. coli*). J Infect Dis 185:1521–1524
35. Reller LB, Weinstein M, Jorgensen JH, Ferraro MJ (2009) Antimicrobial susceptibility testing: a review of general principles and contemporary practices. Clin Infect Dis 49:1749–1755
36. Lin FH, Chen BC, Chou YC, Hsieh CJ, Yu CP (2021) Incidence and risk factors for notifiable typhoid and paratyphoid in Taiwan during the period 2011–2020. Healthcare 9:1316
37. Gupta S, Kashyap B (2016) Bacteriological profile and antibiogram of blood culture isolates from a tertiary care hospital of North India. Trop J Med Res 19:94
38. Velazquez EM, Nguyen H, Heasley KT, Saechao CH, Gil LM, Rogers AW, Bäumler AJ (2019) Endogenous enterobacteriaceae underlie variation in susceptibility to salmonella infection. Nat Microbiol 4:1057–1064
39. Du Z, Su H, Wang W, Ye L, Wei H, Peng Z, Anishchenko I, Baker D, Yang J (2021) The trRosetta server for fast and accurate protein structure prediction. Nat Protoc 16:5634–5651
40. Kresken M, Pfeifer Y, Wagenlehner F, Werner G, Wohlfarth E (2022) Resistance to mecillinam and nine other antibiotics for oral use in *Escherichia coli* isolated from urine specimens of primary care patients in Germany, 2019/20. Antibiotics 11:751
41. Kim C, Latif I, Neupane DP, Lee GY, Kwon RS, Batool A (2021) The molecular basis of extensively drug-resistant *Salmonella typhi* isolates from pediatric septicemia patients. PLoS ONE 16:1–15
42. Zakir M, Khan M, Umar MI, Murtaza G, Ashraf M, Shamim S (2021) Emerging trends of multidrug-resistant (MDR) and extensively drug-resistant (XDR) *Salmonella typhi* in a tertiary care hospital of Lahore, Pakistan. Microorganisms 9:2484–2495
43. Bhumbla U, Chaturvedi P, Jain S (2022) Prevalence of *Salmonella typhi* in among febrile patients in a tertiary care hospital of South West Rajasthan. J Fam Med Prim Care 11:2852–2855
44. Malik N, Ahmed M (2016) In vitro effect of new antibiotics against clinical isolates of *Salmonella typhi*. JCPSP J Coll Physicians Surg Pak 26:288–292
45. Fatima M, Kumar S, Hussain M, Memon NM, Vighio A, Syed MA, Chaudhry A, Hussain Z, Baig ZI, Baig MA, Asghar RJ, Ikram A, Khader Y (2021) Morbidity and mortality associated with typhoid fever among hospitalized patients in Hyderabad District, Pakistan, 2017–2018: retrospective record review. JPHS 7:1–6
46. Du D, Wang-Kan X, Neuberger A, Van-Veen HW, Pos KM, Piddock LJ, Luisi BF (2018) Multidrug efflux pumps: structure, function and regulation. Nat Rev Microbiol 16:523–539
47. Buckner MMC, Blair JMA, Piddock LJV (2016) Emerging resistance to ciprofloxacin and newer antimicrobial agents in *Salmonella enterica* serovar Typhi: a molecular perspective. J Antimicrob Chemother 71:2599–2609
48. Kumar Y, Sharma A, Mani KR (2019) Detection of resistance–nodulation–division (RND) efflux pump genes and their association with multidrug resistance in *Salmonella enterica* serovars isolated from poultry and human clinical samples. Vet World 12:710–715
49. Ziaee M, Javanmard D, Sharifzadeh G, Namaei MH, Azarkar G (2016) Genotyping and mutation pattern in the overlapping MHR region of HBV isolates in Southern Khorasan, Eastern Iran. Hepat Mon 16:37806
50. Hassan MM, Mohamed SB, Hussain MA, Dowd AA (2015) Deleterious nonsynonymous SNP found within HLA-DRB1 gene involved in allograft rejection in Sudanese family: using DNA sequencing and bioinformatics methods. Open J Immunol 5:222
51. Capriotti E, Fariselli P, Casadio R (2005) I-Mutant2. 0: predicting stability changes upon mutation from the protein sequence or structure. Nucleic Acids Res 33:306–310
52. Masso M, Vaisman II (2010) AUTO-MUTE: web-based tools for predicting stability changes in proteins due to single amino acid replacements. Protein Eng Des Sel 23:683–687
53. Zwama M, Yamaguchi A, Nishino K (2022) Recent updates on the structure–function relationship of multidrug efflux pumps. Front Microbiol 13:867084
54. Li D, Zhang L (2022) Structure prediction and potential inhibitors docking of enterovirus 2C proteins. Front Microbiol 13:856574
55. Radhakumar DS, Thiyagarajan S, Rajaram K, Parsanathan R (2023) Human antimicrobial peptide Histatin 1, 3 and its autoproteolytic cleaved peptides target the monkeypox virus surface proteins: molecular modelling and docking studies. Res Sq 18:1–22
56. Xu J (2019) Distance-based protein folding powered by deep learning. Proc Natl Acad Sci USA 116:16856–16865
57. Wang S, Sun S, Li Z, Zhang R, Xu J (2017) Accurate de novo prediction of protein contact map by ultra-deep learning model. PLoS Comput Biol 13:e1005324